Course Code	21EIC553J	Course Name	ROBOT PROGRAMMING	Course Category	C	PROFESSIONAL CORE	L	T	P	C
							3	0	2	4

Pre-requisite Courses	*Nil*	Co- requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*Electronics and Instrumentation Engineering*	Data Book / Codes / Standards	*Nil*		

Course Learning Rationale (CLR):	***The purpose of learning this course is to:***
CLR-1:	*introduces students to foundational concepts in robot programming*
CLR-2:	*covers advanced topics in robot autonomy, emphasizing slam for mapping and localization, techniques*
CLR-3:	*explores topics in multi-robot systems, object tracking, and human-robot interaction*
CLR-4:	*addresses advanced topics in human-robot interaction HRI), dynamic path planning, and advanced kinematics.*

Course Outcomes (CO):	***At the end of this course, learners will be able to:***	***Programme Outcomes (PO)***		
		1	***2***	***3***
CO-1:	*gain proficiency in writing ROS nodes for robot movement, implementing algorithms*	*2*		
CO-2:	*develop the ability to apply SLAM algorithms to create maps and localize robots accurately*	*3*		
CO-3:	*implement object tracking algorithms, and design intuitive HRI interfaces for effective human-robot collaboration.*	*3*	*1*	
CO-4:	*acquire skills in developing interactive interfaces for HRI using voice and gesture recognition, implementing dynamic path planning algorithms*	*3*	*1*	

Module-1 - Basics of Robot Programming and Movement ***15 Hour***

Introduction to ROS, Basic Robot Control, Line Following Algorithm, RoboAnalyzer

Experiments:

1. *Basic movement*
2. *Obstacle avoidance*
3. *Line following robot*

Module-2 - Mapping, Localization, and Navigation ***15 Hour***

SLAM (Simultaneous Localization and Mapping), Localization Techniques, Vision Based Navigations

Experiments:

1. *Mapping and localization*
2. *Autonomous navigation*
3. *Vision based navigation*

Module-3 - Advanced Robot Interaction and Coordination ***15 Hour***

Multi-Robot Systems, Robotic Arm Control, Object Tracking and Following

Experiments:

1. *Multiple robot coordination*
2. *Pick and Place with robotic arm*
3. *Follow the Leader*

Module-4 – Human-Robot Interaction and Advanced Kinematics	**30 Hour**
Human Robot Interaction, Dynamic Path Planning, Advanced Kinematics *Experiments:* *1. Human Robot Interaction using voice and gesture* *2. Path planning in dynamic environment* *3. Implement inverse kinematics for complex robot configuration*	

Learning Resources	1. *Morgan Quigley, Brian Gerkey, and William D. Smart" Programming Robots with ROS", 1st Edition, O'Reily Media, 2015* 2. *Roland Siegwart, Illah R. Nourbakhsh, and Davide Scaramuzza, "Introduction to Autonomous Mobile Robots", 2nd Edition, MIT Press, 2011*	3. *Adrian Kaehler and Gary Bradski, "Learning OpenCV 3: Computer Vision in C++ with the OpenCV Library", 1st Edition, O'Reily Media, 2016* 4. *Mark W. Spong, Seth Hutchinson, and M. Vidyasagar, "Robot Modeling and Control", 1st Edition, Wiley, 2005* 5. *Cameron Hughes, Tracey Hughes, "Robot Programming: A Guide to Controlling Autonomous Robots", Pearson Education, 2016*

Learning Assessment

	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test. (45%)		Life-Long Learning CLA-2 (15%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	-	20%	20%	-
Level 2	Understand	20%	-	-	20%	20%	-
Level 3	Apply	30%	-	-	30%	30%	-
Level 4	Analyze	30%	-	-	30%	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers

Experts from Industry	**Experts from Higher Technical Institutions**	**Internal Experts**
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